

Studor DS Screening Assessment

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1 Executive Summary

The project involves building a framework that can track student engagement and behavior by giving each student a dynamic **engagement score**.

It's quite obvious that students don't disengage suddenly but rather gradually and hence we use this fact to an advantage to identify early on which students are most likely to fail/withdraw.

Using these predictions we are able to alert university staff so they can take precautions and help such students.

PathAI also essentially aims to help university students find the right learning opportunities, and so we additionally create a course recommendation that will help students find suitable learning opportunities.

1. Engagement Scoring System :

- Tracks student engagement dynamically (week-by-week) on a scale of **0-100**
- Used five behavioral signals to do so:
 - Login frequency
 - Recency of activity
 - Number of assessment submissions
 - Activity diversity
 - Engagement trend (increasing or decreasing)
- It captures changes in behavior over time, rather than relying on final outcomes
- Got that students who **passed** had an average score of **34.2** and students who **failed or withdrew** had a lower average score of **27.5**
- This gap shows that the score captures meaningful behavioral differences and can be used as an early indicator of engagement and risk

2. Predictive Disengagement Model :

- Predicts which students are likely to **fail or withdraw** using only **first six weeks of data**
- Enables **early identification**, before the student already withdraws, which is what we wanted.
- It is able to correctly identify **90%** of the at risk students (recall = 0.90)
- It flagged **4,467 students** for advisor attention, and based on the risk score it classifies the students as **HIGH, MEDIUM or LOW** so that the advisor can take suitable action for each case
- The model predictions are well-calibrated, so we can confidently say that when model predicts a **70% risk**, it corresponds to **70% actual failure/withdrawal rate**
- This makes the risk scores **reliable and actionable** for advisors

3. Course Recommendation Engine

- Recommends **three courses** for the student's next semester based on their **academic profile and behavior**
- Using two approaches:
 - **Content-based filtering**: matches a student's profile to courses where similar students performed well (passed)

- **Collaborative filtering:** recommends courses taken by students with similar behavior patterns
- Also handles **new students (cold start)** using a demographic-based strategy
- Ensures every student receives a meaningful recommendation from the start
- Covers **86% of the course catalog**, avoiding repeated recommendations of only popular courses

2 Task 1: Behavioral Scoring Framework

2.1 Feature Engineering and Selection

Started with 8 candidate behavioral features extracted from the VLE clickstream data. Explored each using violin plots and correlation analysis. 3 weakest features were dropped, leaving 5 final features:

- **Active Days** ($r = 0.29$): distinct days logged in per week. Chosen because it captures consistency. Student who log in 5 times each day is better than one who just logs in once for a long time. At-risk students logged in 1-2 days per week and safe students 4-6 days.
- **Recency** ($r = 0.26$): days since last login within the week. Chosen since disengaging students go silent before actually withdrawing.
- **Assessment Submitted** ($r = 0.24$): binary flag. Students who submit lesser assignments are more at risk than those who submit regularly. Violin plot shows the distinction.
- **Total Clicks** ($r = 0.21$): Interesting observation seen here. In violin plot at-risk students have longer tail towards the top, it means that students who are about withdraw/fail are clicking in panic as a last resort before disengaging completely. Therefore it is important to combine with active days and recency.
- **Session Regularity** ($r = 0.19$): standard deviation of daily clicks within a week. Chosen because a lower deviation means more consistent study habits, at-risk students showed more erratic patterns(bursts followed by silence).

2.2 Engagement Score Formula

Each feature was normalized to 0–1 using Min-Max scaling. Recency was inverted (since higher recency = lower engagement). Weighted sum was computed:

$$\text{Score} = 0.30 \times \text{Active Days} + 0.25 \times \text{Recency} + 0.20 \times \text{Submitted} + 0.15 \times \text{Total Clicks} + 0.10 \times \text{Session Regularity}$$

Weights reflect relative correlation strength. The score is multiplied by 100 to produce a 0–100 scale.

2.3 Student Archetypes

Three real students were identified from the data to illustrate distinct engagement trajectories:

- **Steady Engager**: consistently scores 60–85 throughout the semester. Never crashes to zero. Passes with distinction.
- **Early Dropout**: minimal activity from week 0, and activity further declines gradually, on average drop out by week 10.
- **Late Recoverer**: very low engagement in first few weeks, but steadily climbs up from week 10 and catches up with the steady engager. Eventually passes.

2.4 Score Validation

We compared average engagement scores to see the difference. Students who passed averaged **34.2** and those who withdrew/failed averaged **27.5**: a difference of 6.7 points.

The difference shows separation but is not very high because of genuine complexity, most students are a mix of each type of archetype and have quite different behaviour whether they are at risk or safe. Therefore the score works best as an input to the predictive model in Task 2 rather than a standalone threshold.

3 Task 2: Predictive Disengagement Model

3.1 Features: No Data Leakage

Five features aggregated per student strictly within Weeks 0 to 6. They are: **avg weekly clicks**, **avg active days**, **avg session regularity**, **avg recency**, and **total assessment submissions**.

3.2 Model Selection

Logistic Regression outperformed Random Forest (ROC AUC 0.68 vs 0.66, other metrics were also higher for log reg) and hence selected logistic regression. Also logistic regression models tend to be more calibrated naturally.

3.3 Optimizing for Recall

We do this by adjusting the threshold. At the default threshold of 0.50, we got recall as 0.56, which means 44% of at-risk students were missed. This is very bad for us since we do not want to miss getting an alert for a student that is actually at risk.

So we lower the threshold to 0.31, this affects our precision but again missing an at-risk student is much more costlier than getting a false alarm.

lowered the threshold to 0.31 and finally got following:

Precision	Recall	F1	ROC AUC
0.52	0.90	0.66	0.68

4,467 students were flagged — 861 HIGH, 1,435 MEDIUM, 2,171 LOW risk.

3.4 Calibration

The calibration curve confirmed predicted probabilities are reliable, the model's plot was nearly following the $y=x$ line, so a 70% risk score corresponds to approximately 70% actual withdrawal rate, so advisors can trust the numbers they see.

3.5 Top 3 Features

- **Avg Active Days** (−0.45): strongest protective factor. Consistent login days means commitment. Absence leads to withdrawal.
- **Total Submissions** (−0.40): students who stop submitting have behaviorally disengaged before formally withdrawing.
- **Avg Clicks** (+0.12): counterintuitively increases risk. shows panic clicking, frantic activity before disappearing completely as a last resort of trying before they give up.

3.6 Staff Notification

The alert gets triggered for risk score of higher than 31% and based on the risk score we divide the students into 3 risk levels so that the advisor knows who to prioritize and what action to take.

- **HIGH** ($\geq 65\%$): maybe a personal call within 24 hours to understand the student's situation.
- **MEDIUM** (50–65%): probably a personalised check-in email within 48 hours.
- **LOW** (31–50%): just a automated message sent to student on their dashboard, advisor can recheck at Week 8 if their risk score has increased further and take necessary action if so.

4 Task 3: Course Recommendation Engine

4.1 Course Catalog

OULAD contains 7 anonymised modules. Pass rates vary significantly — AAA has the highest pass rate (71%) while CCC has the lowest (38%), suggesting meaningful differences in difficulty and student fit.

4.2 Approach 1: Content Based Filtering

For each module we computed an average profile of students who passed (demographic + behavioral features). When recommending the new student's profile is compared to each course profile using **cosine similarity** and the modules where the new student has most resemblance with the typical successful student is recommended.

Cosine similarity is better over Euclidean distance here because it measures the angle between profile vectors rather than magnitude, because for example two students with similar behavioral patterns but different click volumes are treated as similar, which is appropriate here.

4.3 Approach 2: Collaborative Filtering

We find 150 most similar students using cosine similarity on the full feature matrix (demographics + behavioral history), then recommend modules that those similar students took. So we are using peer interaction patterns rather than course profiles.

4.4 Cold Start Problem

For brand new students with no behavioral history we can match on demographics that would be known during registration (age band and education level).

We created a random example: A student aged 0 to 35 with A Level education. It recommends modules FFF BBB and DDD because it found 5381 similar students and they passed these courses.

4.5 Evaluation

We used **proxy holdout evaluation** : 2,479 students who took 2+ modules were used as the ground truth. Their first module was used as history and we try to predict their second module. When we compare what they actually took and what we predicted we get our metrics.

Method	Precision@3	Coverage
Content-Based	0.22	86%
Collaborative	0.32	86%
Random Baseline	0.50	—

The values are quite low because using this metric isn't very justified because in the OULAD dataset student are picking their second course based on what's available to them or by personal choices or even randomly. So we should not be using this as a method to see how good our recommender is.

Coverage of 86% shows that both methods explore the full catalog rather than just recommending the most popular courses to every student.

5 Limitations and Future Work

5.1 Limitations

- **Engagement score overlap:** the 6.7 point separation between safe and at-risk students is real but modest. Some at-risk students engage heavily before withdrawing and some safe students barely use the VLE therefore the unavoidable overlap.

5.2 Future Work

- **Student risk trajectory:** instead of a single prediction in form of the risk score at week 6 we should show advisors a student's predicted risk score evolving week by week. Then they would be able to tell exactly when disengagement began (when there is sharp rise in risk scores in adjacent weeks), so they can take immediate action at that time.